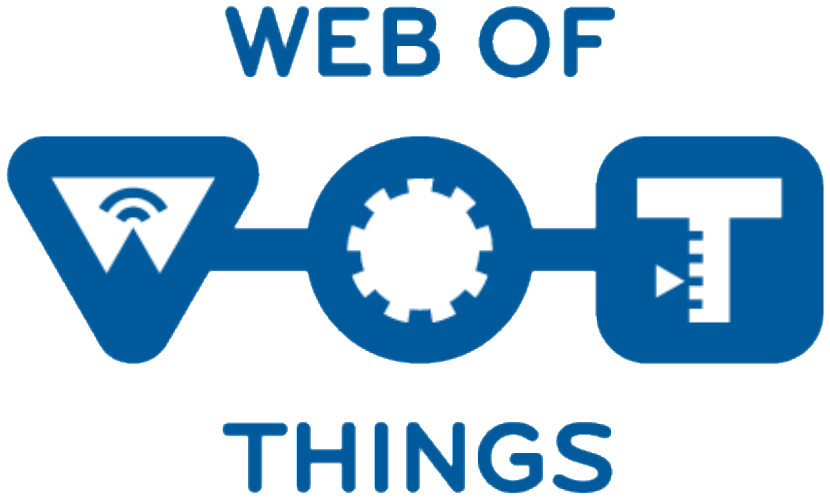


WoT & AI – Physical AI

Brainstorming

Sebastian Käbisch

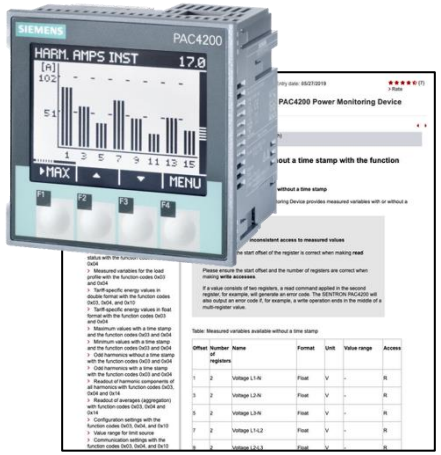
October 29, 2026

TPAC 2026

Goal of the Session

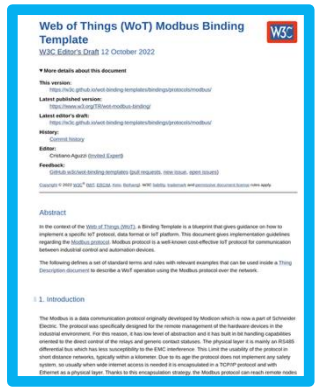
- Precise the usage of WoT & AI
- What are the use cases?
- Plans to publish the deliverable „WoT Primer and Best Practice Deliverable“
- Short presentations from
  - Sebastian
  - Mahda
  - Kunihiko-san

How to create a WoT Thing Description for a device? Manually







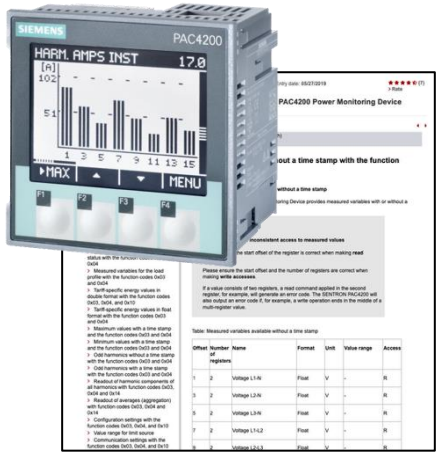


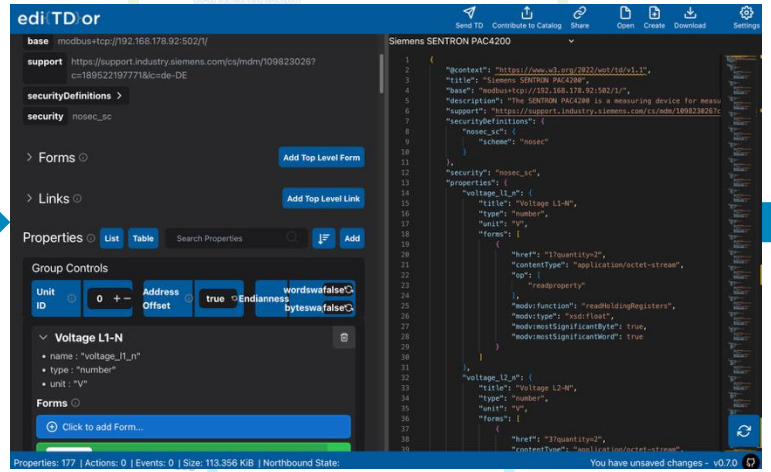
WoT Modbus Binding



- Manually entering or copy&paste information such as from manual into the TD format
- Requires detail knowledge of the WoT standards such as about the Modbus binding
- Efforts needs to be done only once

How to create a WoT Thing Description for a device?
Use TD assistant tools like ediTDor



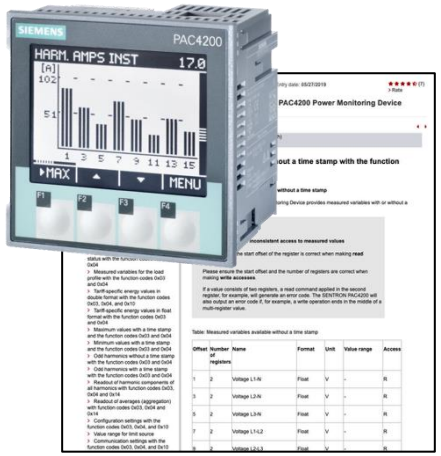


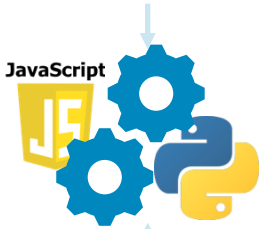


https://eclipse-editdor.github.io/editdor/

- An assistant that helps to create a valid WoT TD step by step
- No detail knowledge of the WoT standards is required
- It is still necessary to provide information about the device interface manually

How to create a WoT Thing Description for a device?
Use transformation scripts (e.g., based on Python or JavaScript)

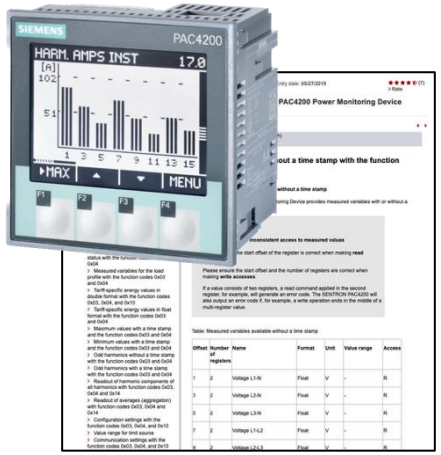


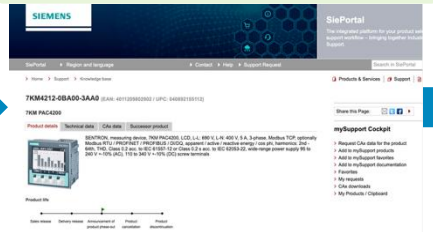



```
{
    "@context": "https://www.w3.org/2022/wot/td/v1.1",
    "base": "modbus+tcp://192.168.10.100:502/1/",
    "support": "https://support.industry.siemens.com/dl/dl-media/595/3
    "securityDefinitions": {
        "nosec_sc": {
            "scheme": "nosec"
        }
    },
    "security": "nosec_sc",
    "properties": {
        "voltage_l1_n": {
            "title": "Voltage L1-N",
            "type": "number",
            "unit": "V",
            "forms": [
                {
                    "href": "40001?quantity=2",
                    "contentType": "application/octet-stream",
                    "modv:function": "readHoldingRegisters",
                    "modv:type": "xsd:float",
                    "modv:mostSignificantByte" : true,
                    "modv:mostSignificantWord" : true,
                }
            ]
        },
        "voltage_l2_n": {
            "title": "Voltage L2-N",
            "type": "number",
            "unit": "V",
            "forms": [
                {
                    "href": "40003?quantity=2",
                    "contentType": "application/octet-stream",
```

- Automized transformation from existing sources (e.g., databases, CVS tables, etc) to create a valid WoT TD
- No detail knowledge of the WoT standards is required
- If the structure of the source changes, a new or updated script is needed

How to create a WoT Thing Description for a device?
Download directly from product page (e.g., via Webpage or TD catalog)

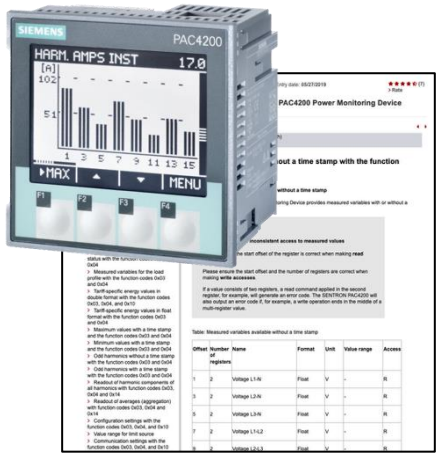


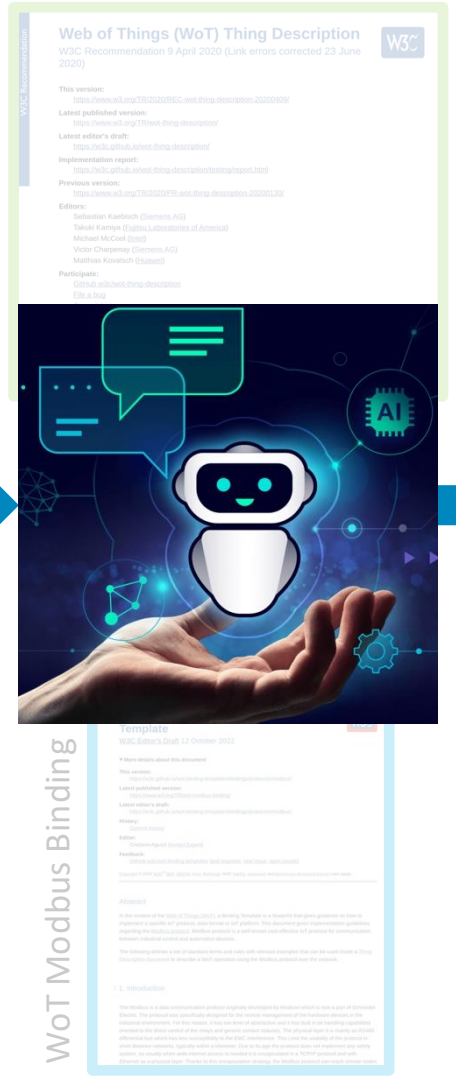




WoT TD direct from manufacturer (highest level of trustworthiness)

How to create a WoT Thing Description for a device? Ask AI






```
{
    "@context": "https://www.w3.org/2022/wot/td/v1.1",
    "base": "modbus+tcp://192.168.10.100:502/1/",
    "support": "https://support.industry.siemens.com/dl/dl-media/595/3
    "securityDefinitions": {
        "nosec_sc": {
            "scheme": "nosec"
        }
    },
    "security": "nosec_sc",
    "properties": {
        "voltage_l1_n": {
            "title": "Voltage L1-N",
            "type": "number",
            "unit": "V",
            "forms": [
                {
                    "href": "40001?quantity=2",
                    "contentType": "application/octet-stream",
                    "modv:function": "readHoldingRegisters",
                    "modv:type": "xsd:float",
                    "modv:mostSignificantByte" : true,
                    "modv:mostSignificantWord" : true,
                }
            ]
        },
        "voltage_l2_n": {
            "title": "Voltage L2-N",
            "type": "number",
            "unit": "V",
            "forms": [
                {
                    "href": "40003?quantity=2",
                    "contentType": "application/octet-stream",
```

- Simple ask for a WoT TD (of any device)
- No detail knowledge of the WoT standards is required
- Trustworthiness; Hallucinations; Costs; Deterministic

WoT Primer and Best Practice Deliverable

- Target audience: developer, engineers, device manufacturers, AI agents

- Content (draft):
  - WoT in a Nutshell (reuse existing material) → WoT Primer ?
  - Getting started (reuse existing material) → WoT Primer ?
    - Your first TD
    - How to talk to devices via TDs
    - Tools
  - Creating Thing Descriptions in Practice
    - Manuell, tools and AI (see previous slides)
      - Providing specific prompts templates
    - Semantic integration
      - @type for grouping
    - How you should use specific terms like „title“ and „description“ in regard to property name
  - WoT and Physical AI
    - TDs enabling AI agents to discover, understand, and operate devices
  - Application Patterns
    - Factory, Building, Smart City, Smart Home, …
    - Device, Edge, Cloud
    - Sleepy devices (caching data values,…)

- We should setup a new TF and do regular meetings (be-weekly)
  - Call for contributors!